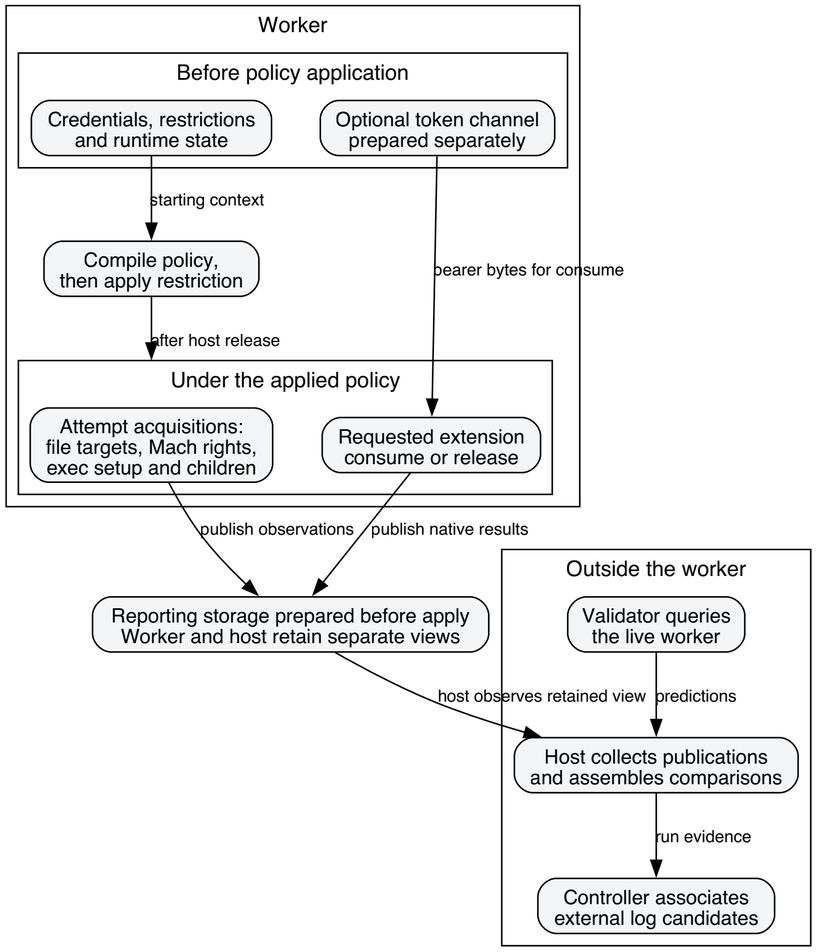
The Worker: Instrument and Constraints

A worker must keep reporting under the policy it is testing while leaving the requested operations exposed to that policy. PolicyWitness prepares its report storage before application, then opens file targets, looks up Mach services and creates exec resources inside the attempts. Moving a target open across that boundary changes the experiment: a later read would exercise an already held descriptor, with the authority that acquiring it supplied.

The boundary is policy application inside a running process. The executable, libraries, memory and process context are already present. A fresh worker applies its policy once: the one worker of the single-worker form, or one worker per actor of the actor form, where an unapplied actor applies none. Its later phases share that process and can observe effects of earlier actions, and in the actor form an actor can hold authority it issued itself. The unsandboxed host arranges queries, collects publications and replies. This tour explains the resources and constraints of that instrument. The architecture tour owns orchestration and ordering; the build tour owns compilation and provenance; the evidence contract owns the meaning of its records.

Capability map



This is a conceptual overview; the detailed capability inventory keeps the individual resources, lifetimes and supporting citations. The diagram draws the declaration form's worker. Actor workers use separate authority regions, one-shot issue/delivery slots and transfer cells, plus per-reference native handles; an unapplied actor proceeds from setup without applying policy. The crosswalk below distinguishes these resources. The exec exclusion is per participating actor, including recipients: another actor in the specimen may exec. Delivery precedes requested consumption and does not consume a token.

Form	Resources and acquisition	Lifetime and ownership
All workers	Evidence object, host/worker mappings, policy and readiness endpoints prepared before setup; fixed request/result storage	The worker retains its evidence descriptor and mapping until exit; the host retains its own view through collection. These setup-held descriptors are reporting apparatus, not fresh target acquisitions.
Declaration	Issuer channel, host delivery duplicate and worker fd 5; borrowed token view and static consume/release handle (S5, S13d, S19–S22, S37)	Issuance precedes worker creation. Accepted bytes remain in the mapped channel until erasure; native handle state persists across actions. Detailed erasure and native-release limits are below.
Actor	Host creates one separate authority region per actor and private custody per token before launch; worker maps fd 5, closes that descriptor and snapshots inbound assignments before application	Host and worker hold distinct views. One-shot outbound/inbound slots and transfer cells cannot be reused. Shared teardown erasure requires a reaped or never-launched owner; unresolved termination permits host detachment only. Private custody has independent erasure ownership.
Actor actions	Native issue allocation, committed outbound slot, host custody copy, assigned inbound slot, per-reference native handle	Worker erases/frees inspected issue allocation before publication; host copies and erases outbound bytes before answering. Recipient framing acceptance and host equality/settlement precede consume. Last-use consume erases inbound bytes even on native failure; ordinary exit erases other settled inbound bytes. Handles persist separately from token bytes.
Requested operations	Fresh file descriptors, Mach rights, exec resources acquired inside the action after setup	These exercise then-current process conditions. Closing a file descriptor, erasing bearer bytes and a native extension release are different operations; later fresh access supplies its own observation.

The actor resource owners are ActorSession, AuthorityCustody and the worker's `prepare_authority` and `erase_authority`. The actor handoff sequence explains their completion gates; exact capacities belong to LIMITS.

The host closes validator collection before releasing the worker to attempt a phase; the worker receives the release ordinal, while predictions stay outside it. The ordering account explains that hold and the evidence needed to establish it.

What the worker can establish

The worker performs admitted actions and publishes native observations. Its inputs include attempt kinds and targets, bounded arguments, phase partitions and time allowances; requested extension actions maintain local handle state. Prediction belongs to the validator. The host owns disposition, collection and comparison, and the controller gathers log candidates. Those responsibilities are a deliberate division of work, not observations the worker failed to make.

The observations themselves have limits. Native return values and local publications do not establish private kernel extension state, effective permissions, revocation or the cause of a permission-shaped failure. Nor do they show whether privacy consent mediation affected an attempt; the external-runner guide gives target-selection guidance for that limitation. Joining host observations and log candidates to a result does not automatically justify causal attribution or a verdict that prediction and enforcement agree. The evidence ownership table and comparison contract define those boundaries.

A native effect and a published result are also distinct. A call can create a file before the worker dies without publishing its result; the host must then withhold that result even though an independent observer can find the file. Native observations and publications explains the publication boundary and its create-then-kill control.

Other omissions are measurement gaps: a live signal-mask observation and a complete initial Mach-right census are absent from the launch evidence, not unknowable.

Reporting apparatus

The host's `ActorSession`, used by both drivers, creates an evidence object, requests its immediate unlink, sizes and maps it, initializes the request and touches its pages. The worker's `map_region` maps the received descriptor before compilation. The mapping holds request data, publication payloads and atomic control fields. The host's mapping remains available during collection even when worker cleanup is unconfirmed. The C harness checks publication under its bare-deny and full-region cases.

The worker retains its evidence descriptor as well as the mapping. Its policy input and readiness descriptors also stay open until exit. Policy EOF ends source delivery and the host closes its readiness reader after the wait, while the worker's references remain. Launch observation L saw those endpoints in both of its recorded launches.

Request, result and token storage have fixed capacities from the limits inventory. The policy buffer and consumed-parameter snapshot are static worker storage. Temporary frames hold timing arithmetic, hash state, argv pointers and diagnostics. Host-side Swift arrays and copied launch strings remain host allocations; spawn copies argument contents into the new image. The worker checks the received source length and bounded fields, then passes the policy and parameter snapshot to libsandbox. Its compiled-object path is `main` → `pw_capture_profile` when requested → application. PW explicitly frees the parameter object and a returned compile-error string on the corresponding paths. The compiled profile has no explicit PW destruction call; ownership changes inside the private apply routine remain unobserved.

The optional compiler receipt uses the already mapped evidence region and reads worker memory before apply. The host `decodeProfileCapture` validates and copies the published receipt. Its relation to compiled bytes, application and effective permissions is defined by the receipt contract.

Starting authority and ambient context

A worker starts with the authority and restrictions of its credentials, signed image and inherited context. Built-in workers carry empty entitlements; external runners supply a separately signed worker, whose read-back entitlements are checked against the specimen's required keys by `enforce_required_entitlements`. The external-runner guide and signing reference own that selection and identity account. Explicit augments change the submitted policy and are recorded as such.

Host spawn actions select the policy, evidence and readiness endpoints, plus the optional authority endpoint. The single-worker launch uses nil spawn attributes and leaves other descriptors to native close-on-exec inheritance. Its standard output/error are inherited, and the evidence descriptor can also survive under its original number if later file actions leave that number intact. The host passes nil envp. The examined Apple spawn wrapper forwards that pointer and the examined exec path constructs an empty environment. In its two recorded empty allow-default launches, L found matching plain and environment-displaying process output, inherited null-device standard output/error, and the expected protocol endpoints. This observation is separate from the source account of inheritance. The contaminated-worker control separately establishes a configuration in which an extra descriptor and environment value reach the worker.

The worker inherits cwd, umask, limits and signal state. It ignores SIGPIPE before compilation. Exec-child setup selects descriptor isolation and a new process group, while leaving ignored-signal inheritance and the signal mask to native spawn rules. Its empty environment and selected standard descriptors have independent inheritance controls. The observer used by those controls itself ignores SIGPIPE, so that observation covers environment and descriptors. For a declaration-form extension request, preparation sets both core limits to zero after resolving the private entry points; local failure is recorded. Public admission in `extensionMeaningFailure` refuses an exec attempt combined with a declaration. Actor normalization excludes exec only in authority-participating actors, including recipients.

Worker launch setup checks file-action initialization, endpoint duplication, evidence unlink and descriptor flags, and spawn attributes before dispatch in `ActorSession`. The host setup controls inject failures in disposable source copies and observe refusal and cleanup. Copied-argument allocation in `withCStringArrayCopy` remains unchecked; failed copies can produce empty or truncated argv. Issuer handle initialization in `ExtensionSession` has fixture-contained invalid-handle use, so its no-spawn result is not proof of production handling of an invalid native handle. An injected unlink failure leaves an owned empty shared-memory name for the fixture to remove.

The single-worker form retains its launch arrangement: the low-descriptor control can allocate evidence at fd 0, which a later policy duplication overwrites. Actor launches relocate sources above their destinations and use close-on-exec by default, explicitly open stdout/stderr on `/dev/null`, and supply an empty environment. A separate kernel descriptor inventory checks actual parked actors with a deliberately inheritable host descriptor and closed host stdin; their explicit endpoints survive and the unrequested descriptors do not.

Launch observation L

The launch-inventory case observed two separate workers with empty allow-default plans, no extensions and no validator queries. Existing pre-ready and post-apply sleep overrides selected the two observation windows. Source order, selected controls and elapsed time locate the observations; the observer did not read publication flags live. Both captures showed numeric descriptors 0–4, read/write null-device stdout/stderr, cwd `/`, and identical plain and environment-displaying `ps` output. `vmmap` returned an inventory; `lsmp` refused task-port inspection. Both workers later published completion and were reaped; the pre-ready worker reported readiness `EPIPE` after the host's readiness wait expired.

These are observations of the recorded app and those two launch configurations. They do not measure a transition within one process, establish an empty Mach-right set or identify an unnamed mapping as the reporting object. Extension-bearing workers, deny-policy acquisitions, live signal state, umask and resource limits were outside the observation.

L's receipt identifies run `20261018T223601Z_85d8418b`, executed from revision `f85c7b094d75bf72931cd4856cb4ad086da20cbe` against the signed app accepted with runtime implementation `a458b28`. The app's own build stamp remains `v0.3.1-39-g6380645-dirty`; its inventory matches that accepted app. Worker source identity is `4cb19db043606894ace2e978434f8805edf4c79245f3588cc68a33b36deca9e8`. The receipt retains the command, specimens, source hashes, selected raw captures, final worker evidence and hashes of the original artifacts. The original complete raw output and app-inventory comparison record are unavailable in this checkout and are not pinned in `tests/RETAINED.json`. The copied observations and hashes describe the dated run; they do not establish current-source applicability.

Reuse for another implementation or environment requires an applicability review or a new observation; matching source hashes establish source identity, not that the relevant conditions were exercised.

Known gap: L covers two recorded launches, and the actor descriptor controls cover specified endpoints and injected descriptor contamination. Live signal masks, resource limits, a complete Mach-right census and private runtime state remain unobserved; native inheritance and loader behavior remain part of the instrument's received substrate, without a complete isolation or effective-authority census.

Acquisitions inside attempts

File targets are opened inside their requested actions. The distinctions among those actions determine what is observed:

Action	Call and retained observation
`attempt_file_open_read`	Opens read-only. The open result determines the published attempt status. On successful open, records an available descriptor-path observation, attempts a one-byte read and close, and discards those latter returns.
`attempt_file_open_write`	Opens with `O_WRONLY | O_TRUNC`. The open result determines the published attempt status. On successful open, attempts to write `x` and close and discards their returns.
`attempt_file_create`	Opens with `O_WRONLY | O_CREAT` and mode argument `0600`, subject to inherited umask. Existing targets are opened without truncation; missing targets can be created. Close is attempted and its return discarded.
`attempt_file_read_fresh`	Opens a fresh close-on-exec descriptor and retains open, metadata, bounded read and single-close results, including partial native evidence when a later call fails.
`write_fresh` / `create_fresh`, through `attempt_file_write_fresh`	Open a fresh descriptor for an existing target or exclusive creation, then retain separate metadata, write and close observations; `write_fresh` also records truncation. Partial native evidence remains when a later call fails.

The existing-file control checks preserved bytes and object identity in its case. The fresh-read contract defines which native observations and bounded bytes reach the reply.

Exec setup is also inside the attempt, after apply. `exec_resources_acquire` creates the output pipes and initializes opaque spawn handles, checks setup results and calls cleanup for initialized resources on failure. The examined Apple spawn implementation allocates both handles and can grow action storage. That is a published-source account; allocation events in the installed libraries were unobserved. PW's post-apply diagnostics also call libc formatting and error-string routines into fixed destinations. The examined formatting implementation and error-string implementation have path-dependent internal behavior, including a retained fallback error buffer. Runtime allocator, locale and binding behavior remain unobserved at these call sites.

The per-attempt resource controls exercise a descriptor ceiling and a bare-deny setup path. The ceiling checks reuse across attempts; the denying case observes setup reaching spawn for its profile and environment. Source establishes post-apply placement. The child receives explicit null-device stdin, captured stdout/stderr, an empty environment and a new process group. PW drains output and observes the leader under the execution budgets. A nonzero child exit after successful spawn still supplies spawn evidence; the exec action contract keeps spawn, child exit and cleanup meanings separate.

`attempt_exec_spawn` retains leader ownership during observation and targets its process group for exceptional cleanup. The optional `exec_observation` records the collection snapshot before cleanup, actual group/fallback call results and the final leader wait separately, behind slot publication. Pending or failed waits have no status; clock faults and local deadlines do not fabricate native errno. Missing metadata establishes no cleanup fact. Worker death can interrupt cleanup; the host supervises the worker. The lifecycle controls observe the exercised leader/descendant paths independently.

Known gap: exec descendants can outlive later steps and worker exit. Ordinary completion requests no termination, so same-group descendants that close their streams can survive; worker death can also leave descendants alive. Exceptional cleanup requests target the owned group, but descendants that leave it escape those requests. Signal returns, EOF and leader reap do not establish group absence.

Mach lookup acquires a bootstrap send-right reference, then a service right on successful lookup. `attempt_mach_lookup` attempts to deallocate the service right and leaves the acquired bootstrap reference until process exit, including after lookup failure and repeated attempts. This lifetime is source-established; a live right-count measurement is absent. `attempt_sysctl_read` uses an attempt-local stack buffer and publishes call status, discarding the returned sysctl value. The inventory includes that buffer among local working storage. Unlink and access calls acquire no PW-owned target handle at their call sites.

Supplied extension authority

The declaration form supplies additional authority deliberately. The host resolves the target and records its metadata, then a fresh ordinary-authority issuer requests a native file token. The issuer and host have their own channel views, separate from worker evidence. Host delivery requires bounded accepted issuance and confirmed clean issuer reaping. `ExtensionSession` owns that preparation and its later host erasure.

The worker's `prepare_extension` resolves consume/release symbols before apply, changes the core limit on the prepared path, maps and validates the optional channel, and attempts to close its descriptor once. Early main refusals can precede that preparation. Accepted token bytes remain a borrowed view of the shared mapping. Requested consume and release calls occur in `attempt_extension`, inside an attempt after apply. The static handle carries state across actions and phases; a native result other than the consume failure sentinel establishes the handle used by a later requested release.

Worker erasure occurs when preparation rejects a mapped channel, after the consume decision on paths reaching that cleanup, before the post-phase test seams, and on every ordinary departure after a channel has been accepted. `erase_accepted_token` clears the full fixed buffer, including bytes beyond the token's terminating NUL. Terminal paths erase before releasing `done`; cancellation and early returns erase without adding publications. Fatal termination or an interrupted native call need not reach cleanup. Erasure does not call native extension release. The accepted mapping remains until exit. The host separately erases its channel before reply preparation and again at session destruction; its observation belongs to the host. In issuer `main`, bounded length inspection governs copying and erasure before free. An oversized token is rejected and that inspected prefix is erased; the allocation extent beyond the bound is unobserved.

Fresh reads obtain a new descriptor at each requested action, so before-consume, during-use and after-release reads each test fresh acquisition at that point. The worker publishes the native release result and any subsequent requested read observations. The extension checks accept either result for the after-release read and verify returned content when successful. The native evidence contract defines the limits of native results and local publication.

In the actor form, `prepare_authority` resolves the issue, consume and release entry points, lowers the core limit, maps the actor's authority region and validates its identity and index before application. Required channel setup failure prevents the scheduler from releasing that actor's phases; missing native symbols instead leave individual operations unavailable. The recipient snapshots each inbound slot's assigned source, action and delivery before application and checks them on publication. `attempt_extension_issue` calls `sandbox_extension_issue_file` once with the exact submitted bytes, publishes nonnull or NULL with the saved errno and copies a bounded token into that issue's assigned outbound slot. It erases the native allocation to its inspected length and frees it before publishing the slot. Published outbound bytes belong to the host, which copies or rejects and erases them; timeout and worker exit do not overwrite bytes a host may still read. `attempt_extension_transfer` publishes into that transfer's own cell and waits for its answer while servicing its own inbound slots, so a transfer to itself cannot wait on itself. Actor consume reads only the inbound slot this actor accepted and the host settled as usable for the named delivery; the handle table is indexed by the plan's handle references rather than one static handle. Parked actors accept or reject deliveries in `service_inbound_once` without consuming. Ordinary exit erases settled inbound buffers. The host erases remaining shared buffers only for reaped or never-launched owners; unresolved termination permits detaching its own mapping, not clearing bytes under a potentially live worker. Private host custody has separate erasure ownership and does not depend on a worker publication.

Local release state differs by form. Declaration execution sets `release_decided` before testing eligibility or making the native call. Actor execution retains its handle-valid bit, but admission rejects a second release of that handle. Neither form provides a public release retry or an automatic exit-time native release. A native release result is separate from later fresh-access evidence.

Known gap: private extension calls may allocate or retain internal state that PW cannot inspect. Erasure observations cover PW-owned transient buffers; successful native results and phase progress do not establish kernel extension state, stable identity, effective permissions or access revocation.

Native observations and publications

A native call can complete before the worker publishes the attempt slot. `pw_progress` records the latest atomic milestone, while `run_attempt` releases the completed slot after writing its payload. The host reads completed payload after observing publication. The create-then-kill control independently observes the created file while requiring the reply to withhold the unpublished result. Partially written payload remains outside the evidence the host is entitled to decode. Native call-boundary flags and result publication have the meanings in the worker evidence contract.

The worker's release wait uses atomic fields, continuous-clock samples and local arithmetic/text. Refusal publishes `done` and exits; cancellation exits without adding `done` or acknowledgement. After the phase loop and any selected post-phase seam, `publish_done_and_exit` releases `done` and calls `_exit(0)` without waiting for the host.

In actor mode the worker also parks before each visit in an activation wait built from the same atomic fields, continuous-clock samples and relax batch as the release wait. It accepts only the global ordinal its next phase record names and publishes that ordinal back; a greater ordinal is a published refusal, an expired shared lifetime cutoff is a published timing failure, and cancellation exits without `done`. A parked actor services inbound transport without running requested native actions; the apparatus cost is one runnable polling loop per parked actor. A cutoff already past at clock setup refuses before any policy byte. An unapplied actor publishes its declared setup after the ready byte and never sets `applied`, and every actor publishes its own hash of the policy bytes it read beside the host's hash of the submitted source. Single-worker requests never enter the activation wait. Public actor requests use this mode, exercised by harness activation controls, actor driver tests and the public request contract. The lifetime cutoff is sampled in checked waits; it does not preempt a blocked native call or bound final cleanup. The host retains its evidence mapping after worker exit and independently observes reaping. Its exit request still cancels a worker waiting for release; grace and termination remain fallback cleanup. These paths use existing storage and automatic frames. Explicit PW dynamic acquisitions after apply are concentrated in the attempted operations; the source-level inventory ends at native/library call boundaries whose internals remain unobserved.

Earlier actions and external actors can change conditions before a later attempt, even with an intact release hold. Unqualified raw query records remain available after a refusal; they do not establish whether each native call encountered a live target. The ordering account and ordering evidence states define eligibility for `query_first` and what an acknowledged release establishes after abnormal exit.

Changes that change the experiment

A proposed improvement can move authority or an observation across a boundary. These examples connect the implementation choice to the question being tested.

Proposed change	Effect on the question	Current behavior and evidence
Open a file target before apply or retain a read descriptor across extension phases	Exercises previously acquired authority	Fresh-read source and extension controls put each acquisition inside its requested action.
Prepare exec pipes and handles before apply	Moves setup outside the specimen's applied conditions	Per-attempt acquisition and resource controls retain that exposure.
Consume a token during preparation or put its bytes in the evidence region	Moves the authority intervention earlier or gives reporting storage bearer authority	Preparation resolves symbols; the separate channel and requested lifecycle calls are covered by the native boundary controls.

Proposed change	Effect on the question	Current behavior and evidence
Retry with a changed target, extra authority or a friendlier policy, or reuse a worker for another specimen	Substitutes different experimental conditions	The driver launches a fresh worker for the submitted specimen. Interrupted I/O and process-observation loops retry within their existing protocols.
Insert `exec_baseline` automatically	Changes the tested policy, including its broad file-read allowance	Augment composition is explicit and recorded in resolved source.
Move reply assembly into the worker	Couples reporting to specimen-controlled communication	The host reads released payload; death before publication preserves an independently observed effect with a withheld result.

Removing the collection hold would also let worker effects change what supposedly prior queries observe. The ordering controls observe both the hold and its limited snapshot meaning.

Detailed capability inventory

The figure and expandable inventory are generated from architecture.json. Each S/N node preserves a resource entry; X nodes represent external evidence. The detailed authority-channel rows describe the declaration form; the crosswalk supplies the actor resources and form differences. This is a resource inventory, not a complete transport sequence. The facts distinguish an object from its descriptors, mappings and borrowed views. The host and worker can map the same object while retaining different references and lifetimes.

Blue nodes are reporting apparatus, including exec reporting resources acquired inside an attempt. Gold nodes represent deliberately supplied authority, restriction or the API/compiler resources used to exercise them. Green nodes are acquisitions by attempted operations. Purple nodes hold evidence assembled outside the worker. White nodes have multiple roles or form the ambient execution substrate: the fresh worker, inherited state, token mappings that also carry metadata, extension state acquired during a requested action, and shared working storage. Starting credentials and policy context remain outside the supplied-authority cluster because part of that context is ambient.

Blue solid edges prepare resources; blue bold edges publish observations. Gold bold edges deliver authority-bearing objects; green edges exercise a resource, including applying a restriction. Purple dotted edges observe and purple dashed edges join evidence. The application boundary is the applied-policy node and the acquisition periods in the inventory. The token channel and reporting mapping are separate. Validator predictions join host evidence outside the worker; the host's release ordinal controls admission to attempts. The edges select relationships among the inventoried resources. Some nodes, including ambient state and temporary storage, stand without a drawn observation path. Open the SVG at its native size to read the complete map.

In the table, H is the host, W the worker, E the host's extension session and I the issuer. P means reporting apparatus, A supplied authority or restriction, and O operation acquisitions. Every use of L in the inventory refers to the two recorded launch configurations, with their stated applicability limits; it does not describe every current worker. IC and CLI are the contaminated-worker and ordinary exec-child controls in the inheritance suite. Acquisition periods record call-site placement, including outputs acquired on compilation return. Check citations name the particular assertion supporting a fact; other implementation details retain source-only support. Policy-specific outcomes belong to their stated specimens and observation conditions. Call-site completeness is established by source review, not by citation checks. Runner guidance describes the review procedure. Process exit reclaims that process's remaining references; created files and references held by other processes have their own lifetimes.

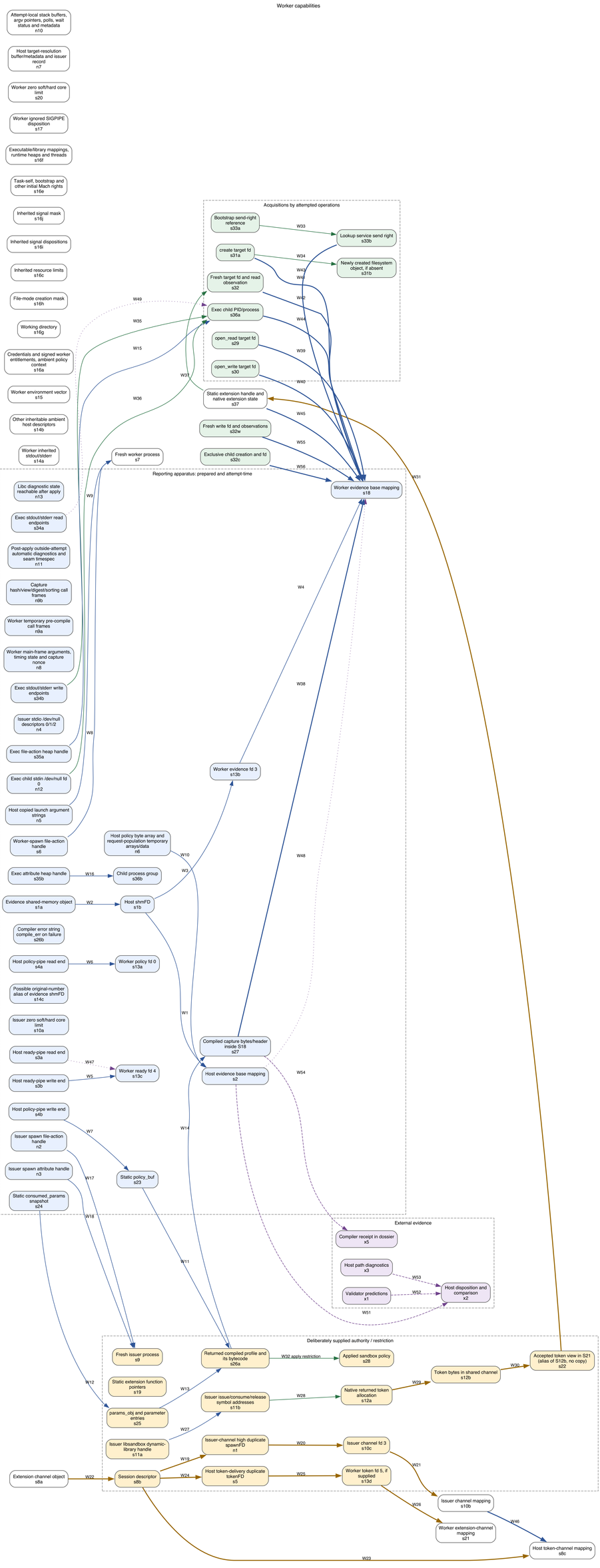

Worker capabilities
Attempt-local stack buffers, argv pointers, polls, wait status and metadata
n10
Host target-resolution buffer/metadata and issuer record
n7
Worker zero soft/hard core limit
s20
Worker ignored SIGPIPE disposition
s17
Executable/library mappings, runtime heaps and threads
s16f
Task-self, bootstrap and other initial Mach rights
s16e
Inherited signal mask
s16j
Inherited signal dispositions
s16i
Inherited resource limits
s16c
File-mode creation mask
s16h
Working directory
s16g
Credentials and signed worker entitlements, ambient policy context
s16a
Worker environment vector
s15
Other inheritable ambient host descriptors
s14b
Worker inherited stdout/stderr
s14a
Acquisitions by attempted operations
Bootstrap send-right reference
s33a
Lookup service send right
s33b
create target fd
s31a
Newly created filesystem object, if absent
s31b
Fresh target fd and read observation
s32
Exec child PID/process
s36a
open_read target fd
s29
open_write target fd
s30
Static extension handle and native extension state
s37
Fresh write fd and observations
s32w
Exclusive child creation and fd
s32c
Fresh worker process
s7
Reporting apparatus: prepared and attempt-time
Worker evidence base mapping
s18
Libc diagnostic state reachable after apply
n13
Exec stdout/stderr read endpoints
s34a
Post-apply outside-attempt automatic diagnostics and seam timespec
n11
Capture hash/view/digest/sorting call frames
n9b
Worker temporary pre-compile call frames
n9a
Worker main-frame arguments, timing state and capture nonce
n8
Exec stdout/stderr write endpoints
s34b
Issuer stdio /dev/null descriptors 0/1/2
n4
Exec file-action heap handle
s35a
Exec child stdin /dev/null fd 0
n12
Host copied launch argument strings
n5
Worker-spawn file-action handle
s6
Worker evidence fd 3
s13b
Host policy byte array and request-population temporary arrays/data
n6
Exec attribute heap handle
s35b
Child process group
s36b
Evidence shared-memory object
s1a
Host shmFD
s1b
Compiler error string compile_err on failure
s26b
Host policy-pipe read end
s4a
Worker policy fd 0
s13a
Possible original-number alias of evidence shmFD
s14c
Issuer zero soft/hard core limit
s10a
Host ready-pipe read end
s3a
Host ready-pipe write end
s3b
Worker ready fd 4
s13c
Host policy-pipe write end
s4b
Issuer spawn file-action handle
n2
Issuer spawn attribute handle
n3
Static policy_buf
s23
Static consumed_params snapshot
s24
Compiled capture bytes/header inside S18
s27
Host evidence base mapping
s2
External evidence
Compiler receipt in dossier
x5
Host path diagnostics
x3
Validator predictions
x1
Host disposition and comparison
x2
Deliberately supplied authority / restriction
Fresh issuer process
s9
Returned compiled profile and its bytecode
s26a
W32 apply restriction
Applied sandbox policy
s28
Accepted token view in S21 (alias of S12b, no copy)
s22
Token bytes in shared channel
s12b
Static extension function pointers
s19
Native returned token allocation
s12a
Issuer issue/consume/release symbol addresses
s11b
params_obj and parameter entries
s25
Issuer libsandbox dynamic-library handle
s11a
Issuer-channel high duplicate spawnFD
n1
Issuer channel fd 3
s10c
Extension channel object
s8a
Session descriptor
s8b
Host token-delivery duplicate tokenFD
s5
Worker token fd 5, if supplied
s13d
Issuer channel mapping
s10b
Worker extension-channel mapping
s21
Host token-channel mapping
s8c
W1
W2
W4
W5
W6
W7
W8
W9
W10
W11
W12
W13
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W56

Figure: worker capabilities. Generated from architecture.json by generate_architecture.py; dot source in worker-capabilities.dot. Figure IDs correspond to the table IDs. Sources support the implementation account. Check citations state their assertion scope; remaining facts have source support only. The generator checks source-symbol presence, resolves cited definitions and requires nonempty scope text; the relevance of each assertion is reviewed, not mechanically established.

Worker capabilities: 84 nodes and 55 edges; source-symbol presence and cited definitions checked

Worker capabilities nodes

Id	Node	Kind	Inventory ID	Acquired by	When	Held across	Release / teardown action	Authority	Reaches exec children	Role	Sources	Checks and assertion scope	Limits
s1a	Evidence shared-memory object	reporting	S1a	H: pw_cworker_shm_open_create, ftruncate	Before compilation	Later phases; last mapping/reference may outlive worker	H requests immediate shm_unlink; injected failure leaves name after teardown	Shared writable input/publication storage	No explicit child reference; see S13b	P	`runCWorker`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
s1b	Host shmFD	reporting	S1b	H: pw_cworker_shm_open_create; checks unlink and descriptor flags (clear in single-worker form, CLOEXEC in actor form)	Before compilation	Through successful worker spawn; host mapping remains through collection	H setup/spawn-failure close, or ownership relinquished and close after transfer deadline starts on successful spawn	Can map/resize the evidence object	Worker receives it; later validators cannot inherit the closed host reference; exec child excludes worker evidence descriptors	P	`runCWorker`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.; `check` — Independent object, descriptor-number reuse and complete-mapping teardown controls; successful worker reporting and reap remain intact.	
s2	Host evidence base mapping	reporting	S2	H: mmap, zero/populate/page-touch	Before compilation	Beyond host descriptor close, through collection including after worker exit or unconfirmed cleanup	H defer munmap	Read worker publications; write input/release/exit request	No: exec replaces address space	P	`runCWorker`	`runCWorkerLifecycleTests` — Controlled worker publications, ownership, acknowledgement reconciliation and confirmed reap; includes acknowledgement recovered after abnormal exit.; `check` — Independent object, descriptor-number reuse and complete-mapping teardown controls; successful worker reporting and reap remain intact.	
s3a	Host ready-pipe read end	reporting	S3a	H: pipe; FD_CLOEXEC	Before compilation	Readiness wait only	H after readiness wait or setup/spawn failure	Observe ready byte	No: close-on-exec	P	`runCWorker`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	

Id	Node	Kind	Inventory ID	Acquired by	When	Held across	Release / teardown action	Authority	Reaches exec children	Role	Sources	Checks and assertion scope	Limits
s3b	Host ready-pipe write end	reporting	S3b	H: pipe; FD_CLOEXEC	Before compilation	Through worker spawn	H immediately after spawn or failure	Supply worker readiness endpoint	Worker fd 4; child excludes it	P	`runCWorker`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
s4a	Host policy-pipe read end	reporting	S4a	H: pipe; FD_CLOEXEC	Before compilation	Through worker spawn	H immediately after spawn or failure	Receive policy bytes	Worker fd 0 replaced in exec child	P	`runCWorker`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
s4b	Host policy-pipe write end	reporting	S4b	H: pipe; F_SETNOSIGPIPE, O_NONBLOCK, FD_CLOEXEC	Before compilation	Through bounded policy transfer	H after transfer or setup/spawn failure	Deliver source and EOF	No: close-on-exec	P	`runCWorker`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
s5	Host token-delivery duplicate tokenFD	authority	S5	H: fcntl(F_DUPFD_CLOEXEC, 64)	Declaration form: before compilation	Through driver return	H defer close	Duplicate authority-bearing channel	Worker fd 5; exec/extension combination refused	A	`runCWorker`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
s6	Worker-spawn file-action handle	reporting	S6	H: posix_spawn_file_actions_init, adddup2	Before compilation	Through driver return	H defer destroy	Prescribes worker descriptor table	Handle no; actions determine worker state	P; setup caveat	`runCWorker`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
s7	Fresh worker process	mixed	S7	H: posix_spawn; single-worker nil attributes/envp; actors explicit endpoints, CLOEXEC_DEFAULT and empty environment	Before compilation	Until exit; unconfirmed cleanup may outlive driver	Terminal publication: erase accepted token, release done, _exit; early return, cancellation, fatal signal or H termination; H waitpid observes reap	Executes specimen with inherited context and signed identity	It parents exec children; not itself inherited	none (ambient execution substrate)	`runCWorker`; `publish_done_and_exit`; `ActorSession`	`launch_inventory` — Requires identified worker captures, readiness and reap/absence; specific inventory contents are the separately identified observation L.; `terminal_exit_without_request` — Applied worker publishes done and exits cleanly without a host exit request.	

Id	Node	Kind	Inventory ID	Acquired by	When	Held across	Release / teardown action	Authority	Reaches exec children	Role	Sources	Checks and assertion scope	Limits
s8a	Extension channel object	mixed	S8a	E: pw_cworker_shm_open_create, ftruncate	Before compilation	Issuer then worker; through session	E requests shm_unlink; injected failure leaves name after teardown	Holds transferable token and issuer metadata	No admitted exec+extension plan	A + E	`ExtensionSession`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.; `check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
s8b	Session descriptor	authority	S8b	E: pw_cworker_shm_open_create, FD_CLOEXEC	Before compilation	Through session	E deinit close, including setup-failure returns	Map/duplicate token channel	No: close-on-exec except explicit delivery dup	A	`ExtensionSession`	`check` — Instrumented issuer-session setup controls observe descriptor cleanup; shared-object identity is not established by matching descriptor metadata.	
s8c	Host token-channel mapping	mixed	S8c	E: mmap, zero/header/path/authority writes	Before compilation	Through reply preparation	E erase before reply; deinit erases, unmaps	Read issuance evidence; supply/erase token	No: address space replaced	A + E	`ExtensionSession`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
s9	Fresh issuer process	authority	S9	E: posix_spawn with CLOEXEC_DEFAULT, nil envp	Before compilation	Until issuer exit; delivery requires confirmed clean reap	I return; E wait/terminate	Ordinary signed issuer authority for file token issuance	No: sibling, reaped before worker delivery	A	`ExtensionSession`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
s10a	Issuer zero soft/hard core limit	reporting	S10a	I: setrlimit(RLIMIT_CORE)	Before compilation	Until issuer exit	Exit; failure returns before mapping	Restricts core-dump production	No: issuer creates no exec child	P	`main`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
s10b	Issuer channel mapping	mixed	S10b	I: mmap from fd 3	Before compilation	Until publication/exit	Normal munmap; malformed header exits without explicit unmap	Write token and issuance metadata	No child	A + E	`main`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	

Id	Node	Kind	Inventory ID	Acquired by	When	Held across	Release / teardown action	Authority	Reaches exec children	Role	Sources	Checks and assertion scope	Limits
s10c	Issuer channel fd 3	authority	S10c	E adddup2, received by I	Before compilation	Through I mapping attempt	I close(3) after mmap; earlier errors leave to exit	Map channel	No child	A	`ExtensionSession`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
s11a	Issuer libsandbox dynamic- library handle	authority	S11a	I: dlopen(RTLD_NOW | RTLD_LOCAL)	Before compilation	Until exit; no explicit dlclose	Loader/process teardown; internals unobserved	Makes private issuance API callable, not an entitlement grant	No child	A	`main`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
s11b	Issuer issue/ consume/release symbol addresses	authority	S11b	I: three dlsym calls	Before compilation	Issuance check/call; loader image remains	Automatic values end with I; no independent handle release	API entry points; consume/release are checked, not invoked	No child	A	`main`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
s12a	Native returned token allocation	authority	S12a	I: sandbox_extension_issue_file	Before compilation	Copy/reject decision only	I bounded erase then free; oversized tail caveat	Bearer bytes for internally selected native read or read-write extension consumption	No: issuer storage not inherited	A	`main`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
s12b	Token bytes in shared channel	authority	S12b	I: memcpy on bounded accepted token	Before compilation	Until worker consume decision or ordinary departure after channel acceptance; earlier mapping/ header refusals leave erasure to the host; host erasure is independent; fatal interruption may bypass worker cleanup	Worker full-buffer erasure in S22; independent host erasure in S8c	Transferable extension token	No admitted exec+extension plan	A	`main`; `erase_accepted_token`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.; `terminal_erasure` — Independent full-capacity and publication-boundary observations reject shortened or late erasure, including refusal paths.	
s13a	Worker policy fd 0	reporting	S13a	H adddup2(policyPipe[0], 0); W default policy fd	Before compilation	Later attempts and phases, until exit	No W close; exit	Read remaining policy stream; normally EOF after source receipt	No: child addopen replaces stdin	P, retained after use	`runCWorker`	`launch_inventory` — Requires identified worker captures, readiness and reap/ absence; specific inventory contents are the separately identified observation L.	

Id	Node	Kind	Inventory ID	Acquired by	When	Held across	Release / teardown action	Authority	Reaches exec children	Role	Sources	Checks and assertion scope	Limits
s13b	Worker evidence fd 3	reporting	S13b	H requests adddup2(shmFD, 3); low-fd collision can substitute policy pipe	Before compilation	Later attempts and phases, until exit	No W close; exit	Map backing object separately from existing mapping	No: child CLOEXEC_DEFAULT	P	`runCWorker`	`launch_inventory` — Requires identified worker captures, readiness and reap/absence; specific inventory contents are the separately identified observation L.; `check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
s13c	Worker ready fd 4	reporting	S13c	H requests adddup2(readyPipe[1], 4); self-dup control leaves fd absent	Before compilation	Later attempts and phases, until exit	No W close; exit	Write readiness pipe; host later closes reader	No: child CLOEXEC_DEFAULT	P	`runCWorker`	`launch_inventory` — Requires identified worker captures, readiness and reap/absence; specific inventory contents are the separately identified observation L.; `check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
s13d	Worker token fd 5, if supplied	authority	S13d	H checks adddup2(tokenFD, 5) before spawn	Declaration form: before compilation	Through extension preparation	prepare_extension closes once; early main refusals leave to exit	Map token channel	No admitted exec+extension plan; child flags would exclude it	A	`runCWorker`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.; `check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
s14a	Worker inherited stdout/stderr	mixed	S14a	H spawn, without replacement file actions	Before compilation	Until exit	Exit; no explicit W close	Whatever underlying endpoints allow; L saw read/write /dev/null	No: exec-child capture pipes replace 1/2	none	`runCWorker`	`launch_inventory` — Requires identified worker captures, readiness and reap/absence; specific inventory contents are the separately identified observation L.	

Id	Node	Kind	Inventory ID	Acquired by	When	Held across	Release / teardown action	Authority	Reaches exec children	Role	Sources	Checks and assertion scope	Limits
s14b	Other inheritable ambient host descriptors	mixed	S14b	Single-worker H spawn without CLOEXEC_DEFAULT; actor launch excludes unrequested descriptors	Before compilation	Until exit unless a file action overwrites number	No generic W sweep; exit	Depends on inherited objects	No: exec-child CLOEXEC_DEFAULT; IC/CLI test child isolation	none	`runCWorker`; `POSIX_SPAWN_CLOEXEC_DEFAULT`	`contaminated_worker` — Independent observer receives the canary descriptor/ environment in the worker adapter and excludes them from exec children; checks child stdio and stdin EOF.	
s14c	Possible original-number alias of evidence shmFD	reporting	S14c	Single-worker form: H clears FD_CLOEXEC and duplicates to 3 without closing the child source alias; actor form relocates and closes original aliases at exec	Before compilation	Until worker exit if original number survives file actions; closing the host descriptor does not remove this child reference	No generic W sweep; exit	Same evidence object as S13b, not a second object	No: exec-child CLOEXEC_DEFAULT	P	`runCWorker`	Source only; no assertion cited	
s15	Worker environment vector	mixed	S15	H posix_spawn(..., nil)	Before compilation	Until exit	Process teardown	No entries observed; potential environment-dependent behavior is source-scoped	No: explicit {NULL} child env; IC/CLI test this boundary	none	`runCWorker`	`launch_inventory` — Requires identified worker captures, readiness and reap/ absence; specific inventory contents are the separately identified observation L.	
s16a	Credentials and signed worker entitlements, ambient policy context	mixed	S16a	Kernel/loader at H spawn; controller checks required keys	Before compilation	Until exit; SBPL/ extension interventions alter effective access	Exit, not a PW reset	Starting access and restrictions; built-in manifest has empty entitlements	Credentials/policy inherit subject to exec semantics; child signature supplies own entitlements	Supplied entitlements plus ambient authority/restrictions; mixed starting substrate	`runCWorker`; `enforce_required_entitlements`	Source only; no assertion cited	
s16g	Working directory	mixed	S16g	Received at H spawn; no PW reset	Before compilation	Later attempts/ phases, until exit	Exit	Relative file-target resolution	Yes, no chdir action	none	`runCWorker`	`launch_inventory` — Requires identified worker captures, readiness and reap/ absence; specific inventory contents are the separately identified observation L.	
s16h	File-mode creation mask	mixed	S16h	Received at H spawn; no PW reset	Before compilation	Later attempts/ phases, until exit	Exit	Reduces requested creation mode	Yes, no umask action	none	`runCWorker`	Source only; no assertion cited	
s16c	Inherited resource limits	mixed	S16c	Received at H spawn	Before compilation	Until exit, except extension core-limit intervention	Exit; S20 changes core limit	Caps allocation, files, core, processes etc.	Yes by spawn inheritance; public extension/exec exclusion applies	none	`runCWorker`	Source only; no assertion cited	
s16i	Inherited signal dispositions	mixed	S16i	Received at H spawn, nil attributes	Before compilation	Until exit except S17	W changes SIGPIPE only	Affects fatal delivery and ignored signals	Ignored dispositions persist, caught handlers reset by exec	none	`runCWorker`	Source only; no assertion cited	
s16j	Inherited signal mask	mixed	S16j	Received at H spawn, nil attributes	Before compilation	Until exit	No PW reset	Controls blocked signal delivery	Yes, no SETSIGMASK	none	`runCWorker`	Source only; no assertion cited	
s16e	Task-self, bootstrap and other initial Mach rights	mixed	S16e	Kernel/runtime initialization at spawn	Before compilation	Until exit or library changes	No PW general release; full live right census unavailable	Task/IPC authority, bootstrap namespace access	Special-port inheritance/native exec rules; exact set unknown	none	`runCWorker`	`launch_inventory` — Requires identified worker captures, readiness and reap/ absence; specific inventory contents are the separately identified observation L.	

Id	Node	Kind	Inventory ID	Acquired by	When	Held across	Release / teardown action	Authority	Reaches exec children	Role	Sources	Checks and assertion scope	Limits
s16f	Executable/library mappings, runtime heaps and threads	mixed	S16f	Loader/runtime creates new image at H spawn; later library calls	Before compilation initially; later internal changes unknown	Until exit or library release	Native runtime; no PW census/teardown	Executable code, storage and runtime service access	Not the same mappings/threads: exec loads a new image	none	`runCWorker`	`launch_inventory` — Requires identified worker captures, readiness and reap/absence; specific inventory contents are the separately identified observation L.	
s17	Worker ignored SIGPIPE disposition	mixed	S17	W: signal(SIGPIPE, SIG_IGN)	Before compilation	All attempts/phases, until exit	No reset	Survive broken ready-pipe write; also changes exec-child behavior	Yes: no SETSIGDEF; no dedicated live signal control	P + effect on O	`main`	`launch_inventory` — Pre-ready hold requires readiness errno EPIPE followed by clean worker completion and confirmed reap.	
s18	Worker evidence base mapping	reporting	S18	map_region: fstat, mmap(MAP_SHARED)	Before compilation	Later attempts/phases, until exit	No explicit W unmap; exit	Read requests, publish results, observe release/exit	No: exec replaces mappings and excludes fd 3	P	`map_region`	Source only; no assertion cited	
s19	Static extension function pointers	authority	S19	prepare_extension: dlsym(RTLD_DEFAULT)	Declaration form: before compilation	Later attempts/phases, until exit	Static storage at exit; no dynamic handle acquired here	Callable native consume/release, no pre-apply consumption	No admitted combination; pointer storage not inherited across exec	A	`prepare_extension`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
s20	Worker zero soft/hard core limit	mixed	S20	prepare_extension: setrlimit(RLIMIT_CORE) after symbols resolve	Declaration form: before compilation	All later attempts/phases, until exit if call succeeds	No reset; exit	Restricts core dumping; missing symbols/setup failure qualify it	No admitted exec with extension; hypothetical ABI child would inherit	P + process context	`prepare_extension`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
s21	Worker extension-channel mapping	mixed	S21	prepare_extension: mmap(MAP_SHARED)	Declaration form: before compilation	Accepted mapping until exit, including after erase	Rejected mapping: erase/munmap; accepted: no explicit unmap	Read/erase token and inspect channel admission metadata	No admitted combination; address space replaced on exec	A + P	`prepare_extension`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
s22	Accepted token view in S21 (alias of S12b, no copy)	authority	S22	prepare_extension: length/NUL/header inspection	Declaration form: before compilation	Until consume decision or ordinary departure after acceptance; fatal interruption may bypass cleanup	Full-buffer erase before done, early return or cancellation exit; no native release; separate host erasure remains	Consumable bearer token; validation does not consume it	No admitted combination	A	`prepare_extension`; `erase_accepted_token`; `publish_done_and_exit`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.; `terminal_erasure` — Independent full-capacity and publication-boundary observations reject shortened or late erasure, including refusal paths.	

Id	Node	Kind	Inventory ID	Acquired by	When	Held across	Release / teardown action	Authority	Reaches exec children	Role	Sources	Checks and assertion scope	Limits
s23	Static policy_buf	reporting	S23	W static storage; read_all_from_fd fills it	Before compilation	Later attempts/ phases, until exit	No wipe or explicit release	Policy input, not a target-file descriptor	No: new exec image	P	`main`	`bare_deny_default` — The direct C harness observes applied policy, terminal publication and a denied file-open attempt under bare deny-default; host Swift handling is outside this control.	
s24	Static consumed_params snapshot	reporting	S24	W: memcpy from bounded shared parameters	Before compilation	Later attempts/ phases, until exit	No wipe; static storage teardown	Exact compile/capture inputs; no additional access	No: new exec image	P	`main`	`params_round_trip` — Named TARGET parameter reaches the compiled policy: the configured path attempt returns EPERM after successful apply; allocation/free details remain source-only.	
s25	params_obj and parameter entries	authority	S25	W: sandbox_create_params, sandbox_set_param	Before compilation	Until compile return, or parameter refusal	sandbox_free_params on success/failure paths after creation	Supplies named compiler inputs	No: freed before possible child spawn	A	`main`	`params_round_trip` — Named TARGET parameter reaches the compiled policy: the configured path attempt returns EPERM after successful apply; allocation/free details remain source-only.	
s26a	Returned compiled profile and its bytecode	authority	S26a	W: sandbox_compile_string	On compilation return, before apply	Passed to capture/ apply; allocation lifetime inside apply unknown	No explicit sandbox_free_profile; exit reclaims remaining process storage	Input to restriction application; not an applied policy by itself	No: new exec image; applied restriction separate S28	A	`main`	`bare_deny_default` — The direct C harness observes applied policy, terminal publication and a denied file-open attempt under bare deny-default; host Swift handling is outside this control.	
s26b	Compiler error string compile_err on failure	reporting	S26b	W: sandbox_compile_string out parameter	On compilation return, before apply	Failure reporting only	W free when non-null and profile null	None; diagnostic bytes	No child on failed compile path	P	`main`	`compile_failure` — Malformed source publishes compile refusal without apply or attempts and exits cleanly; native error allocation/free is source-only.	
s27	Compiled capture bytes/header inside S18	reporting	S27	pw_capture_profile / pw_read_memory: mach_vm_read_overwrite, hashes	Between compilation and apply	Later attempts/ phases, until exit; host copies accepted receipt	No separate allocation/unmap; S18 teardown	Read own task memory via preexisting task port; publish receipt	No: new image, evidence fd excluded	P	`pw_capture_profile`	`max_targets_reply_survives` — The maximal-metadata workload requires a captured compiled-profile receipt with positive bytecode length in the returned reply.	
s28	Applied sandbox policy	authority	S28	W: sandbox_apply(profile)	Between compilation and apply closing boundary	All later attempts/ phases, until exit on success	One-way, no undo	Restricts starting authority; extension may augment access	Inherited sandbox constraints	A	`main`	`bare_deny_default` — The direct C harness observes applied policy, terminal publication and a denied file-open attempt under bare deny-default; host Swift handling is outside this control.	

Id	Node	Kind	Inventory ID	Acquired by	When	Held across	Release / teardown action	Authority	Reaches exec children	Role	Sources	Checks and assertion scope	Limits
s29	open_read target fd	operation	S29	attempt_file_open_read: open(O_RDONLY)	After apply within an attempt	Attempt only on ordinary cleanup path	close after best-effort one-byte read; return ignored	Read opened object	No: closed before later exec, which also excludes it	O	`attempt_file_open_read`	`bare_deny_default` — The direct C harness observes applied policy, terminal publication and a denied file-open attempt under bare deny-default; host Swift handling is outside this control.	
s30	open_write target fd	operation	S30	attempt_file_open_write: open(O_WRONLY | O_TRUNC)	After apply within an attempt	Attempt only on ordinary cleanup path	close after unchecked one-byte write	Write/truncate opened object; actual write result unpublished	No: later exec excludes it	O	`attempt_file_open_write`	`file_actions_have_exact_effects` — Independent file bytes establish the open-write/truncation effect; the worker does not publish the write return.	
s31a	create target fd	operation	S31a	attempt_file_create: open(O_WRONLY | O_CREAT, 0600)	After apply within an attempt	Attempt only on ordinary cleanup path	close, return ignored	Write existing object or create missing object	No: later exec excludes it	O	`attempt_file_create`	`create_existing_file_preserves_contents` — Pre-existing targets retain their bytes after create-open; this does not exercise absent-target creation.	
s31b	Newly created filesystem object, if absent	operation	S31b	attempt_file_create: same open	After apply within an attempt	Later attempts/phases and beyond worker exit	No automatic unlink; later action/external actor may remove	Persistent filesystem effect, mode subject to umask	Child can resolve it by path, not inherited fd	O	`attempt_file_create`	`create_allow` — An initially absent target exists after the successful create attempt under allow-default; the worker also publishes its observed path.	
s32	Fresh target fd and read observation	operation	S32	attempt_file_read_fresh: open(O_RDONLY | O_CLOEXEC), fstat, read	After apply within an attempt	Attempt only on successful close; close failure retained	One close, result published; no retry	Read and identify fresh opened object	No: close-on-exec and child exclusion	O	`attempt_file_read_fresh`	`extensions` — Fresh-read and extension action evidence is compared with independent file bytes and native-call controls; native release is not a revocation assertion.	
s32w	Fresh write fd and observations	operation	S32w	attempt_file_write_fresh: open(O_WRONLY | O_CLOEXEC), fstat, ftruncate, write	After apply within an attempt	Attempt only on successful close; close failure retained	One close; failure retained; completed bytes may persist	Modify the opened existing object	No: close-on-exec and child exclusion	O	`attempt_file_write_fresh`	`fresh_write` — Production syscall substitutions assert open/stat/truncate/write/close order, native failures, short writes and combined-failure preservation; no native policy effect is inferred.; `extension_directory` — Owned existing files and initially absent children provide independent contents and entry observations through ordinary CLI phases, with authority controls and publication withholding.	

Id	Node	Kind	Inventory ID	Acquired by	When	Held across	Release / teardown action	Authority	Reaches exec children	Role	Sources	Checks and assertion scope	Limits
s32c	Exclusive child creation and fd	operation	S32c	attempt_file_write_fresh: lstat, open(O_WRONLY | O_CREAT | O_EXCL | O_CLOEXEC, 0600), fstat, write	After apply within an attempt	Attempt only on successful close; close failure retained	One close; failure retained; created entry and bytes may persist	Create an absent child and write its opened object	No: close-on-exec and child exclusion	O	`attempt_file_write_fresh`	`fresh_write` — Production syscall substitutions assert open/stat/truncate/write/close order, native failures, short writes and combined-failure preservation; no native policy effect is inferred.; `extension_directory` — Owned existing files and initially absent children provide independent contents and entry observations through ordinary CLI phases, with authority controls and publication withholding.	
s33a	Bootstrap send-right reference	operation	S33a	attempt_mach_lookup: task_get_special_port	After apply within an attempt	Later attempts/phases, until exit after successful acquisition	No matching mach_port_deallocate, even when lookup fails	Bootstrap namespace request authority	No explicit transfer; special bootstrap port inheritance is separate	O, unintended retention	`attempt_mach_lookup`	Source only; no assertion cited	
s33b	Lookup service send right	operation	S33b	attempt_mach_lookup: bootstrap_look_up	After apply within an attempt	Attempt on successful lookup	mach_port_deallocate(mach_task_self, svc), result ignored	Send to looked-up service	No explicit port-transfer action	O	`attempt_mach_lookup`	Source only; no assertion cited	
s34a	Exec stdout/stderr read endpoints	reporting	S34a	exec_resources_acquire: two pipe calls	After apply within an attempt	One attempt	exec_resources_release, including partial setup/spawn failure	Read child output	No: explicit addclose and CLOEXEC_DEFAULT	P	`exec_resources_acquire`	`exec_pipes_per_attempt` — Repeated successful execs and surrounding reads under a hard descriptor ceiling constrain descriptor accumulation across attempts; call placement remains source-derived.; `contaminated_worker` — Independent observer receives the canary descriptor/environment in the worker adapter and excludes them from exec children; checks child stdio and stdin EOF.	
s34b	Exec stdout/stderr write endpoints	reporting	S34b	exec_resources_acquire: two pipe calls	After apply within an attempt	Until successful spawn in worker; child lifetime may extend	Worker closes after spawn; setup/spawn failure releases; child exit closes own refs	Child output channel	Yes, duplicated as 1/2; originals explicitly closed	P	`exec_resources_acquire`	`exec_pipes_per_attempt` — Repeated successful execs and surrounding reads under a hard descriptor ceiling constrain descriptor accumulation across attempts; call placement remains source-derived.; `contaminated_worker` — Independent observer receives the canary descriptor/environment in the worker adapter and excludes them from exec children; checks child stdio and stdin EOF.	
s35a	Exec file-action heap handle	reporting	S35a	exec_resources_acquire: init, addopen/dup2/close	After apply within an attempt	One attempt	exec_resources_release destroys initialized handle	Select child descriptors, including /dev/null	Handle no; action effects yes	P	`exec_resources_acquire`	Source only; no assertion cited	

Id	Node	Kind	Inventory ID	Acquired by	When	Held across	Release / teardown action	Authority	Reaches exec children	Role	Sources	Checks and assertion scope	Limits
s35b	Exec attribute heap handle	reporting	S35b	exec_resources_acquire: init/setflags/setpgroup	After apply within an attempt	One attempt	exec_resources_release destroys initialized handle	Child fd isolation and process-group choice	Handle no; flags/group effects yes	P	`exec_resources_acquire`	Source only; no assertion cited	
s36a	Exec child PID/process	operation	S36a	attempt_exec_spawn: posix_spawn	After apply within an attempt	Attempt normally; unconfirmed reap or worker death can leave child alive	Wait observation, conditional group kill, bounded nonblocking reap	Execute requested target under inherited sandbox	This is the child; it can spawn descendants	O	`attempt_exec_spawn`	`exec_pipes_per_attempt` — Repeated successful execs and surrounding reads under a hard descriptor ceiling constrain descriptor accumulation across attempts; call placement remains source-derived.; `contaminated_worker` — Independent observer receives the canary descriptor/environment in the worker adapter and excludes them from exec children; checks child stdio and stdin EOF.; `closed_streams_leader_exit` — Independent peer identity/group observations establish same-group survival after ordinary completion, then fixture cleanup.; `worker_dies_during_exec` — Independent child observations establish survival after worker death and retain fixture-owned cleanup.; `test_production` — Production exec code with controlled native returns checks collection, termination-request and leader-reap separation and native call order.	
s36b	Child process group	reporting	S36b	Exec spawn attributes SETPGROUP, pgroup 0	After apply within an attempt	As long as members remain, possibly beyond attempt/worker	No request on ordinary completion; exceptional group kill while leader ownership is known; no group-absence claim	Cleanup addressing for members that remain	Yes to descendants unless they leave group	P	`exec_resources_acquire`; `attempt_exec_spawn`	`closed_streams_leader_exit` — Independent peer identity/group observations establish same-group survival after ordinary completion, then fixture cleanup.; `worker_dies_during_exec` — Independent child observations establish survival after worker death and retain fixture-owned cleanup.; `test_production` — Production exec code with controlled native returns checks collection, termination-request and leader-reap separation and native call order.	

Id	Node	Kind	Inventory ID	Acquired by	When	Held across	Release / teardown action	Authority	Reaches exec children	Role	Sources	Checks and assertion scope	Limits
s37	Static extension handle and native extension state	mixed	S37	attempt_extension: consume result other than -1	Declaration form: after apply within an attempt	Across attempts/ phases; requested release is not proven revocation	Declaration release_decided set before eligibility/native call; explicit release once if eligible; no automatic exit-time release call; OS teardown	Native extension access represented by handle; exact kernel state unobserved	No admitted exec+extension plan	A + O	`attempt_extension`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
n1	Issuer-channel high duplicate spawnFD	authority	N1	E: fcntl(F_DUPFD_CLOEXEC, 64)	Before compilation	Through E initializer	E defer close; fd 3 duplicate lives in issuer	Selects channel from a high source; low-fd control preserves delivery	No: issuer sibling, duplicate close-on-exec	A	`ExtensionSession`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.; `check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
n2	Issuer spawn file-action handle	reporting	N2	E: init, adddup2, three addopen	Before compilation	Through E initializer	E defer destroy	Channel/stdio selection	No worker memory inheritance	P	`ExtensionSession`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.; `check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
n3	Issuer spawn attribute handle	reporting	N3	E: init and setflags(CLOEXEC_DEFAULT)	Before compilation	Through E initializer	E defer destroy	Issuer fd isolation	No worker memory inheritance	P	`ExtensionSession`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.; `check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
n4	Issuer stdio /dev/null descriptors 0/1/2	reporting	N4	E: posix_spawn_file_actions_addopen, executed by native spawn	Before compilation	Issuer lifetime	Issuer exit	Read EOF/write discard	No: issuer creates no child	P	`ExtensionSession`	Source only; no assertion cited	

Id	Node	Kind	Inventory ID	Acquired by	When	Held across	Release / teardown action	Authority	Reaches exec children	Role	Sources	Checks and assertion scope	Limits
n5	Host copied launch argument strings	reporting	N5	withCStringArrayCopy : strdup for H/E invocations	Before compilation	Spawn call only	Helper defer free; Swift array lifetime	Transfer launch arguments, not token bytes	Arguments copied into new images, host pointers not inherited	P; setup caveat	`withCStringArrayCopy`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
n6	Host policy byte array and request-population temporary arrays/data	reporting	N6	H: Array(input.policy.utf8), writeString, captureNonceBytes	Before compilation	Transfer/population scope; input objects may live through run	Swift scope/ARC, not worker teardown	Exact delivered inputs and nonce	No host memory inheritance	P	`runCWorker`	Source only; no assertion cited	
n7	Host target-resolution buffer/metadata and issuer record	mixed	N7	E: [CChar], caller-buffer realpath, fstatat, strings/record	Before compilation	Through preparation; record survives into reply	Swift scope/ARC	Resolve token target before worker starts; metadata is host evidence	No	A + E	`ExtensionSession`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
n8	Worker main-frame arguments, timing state and capture nonce	reporting	N8	W local declarations, pw_clock_start, nonce memcpy	Before compilation	Main frame until exit; timing changes across phases	Process exit (main never returns after successful apply)	Operate release/ timing protocol; no new access handle	No: new exec image	P	`main`	Source only; no assertion cited	
n9a	Worker temporary pre-compile call frames	reporting	N9a	parse_args, map_region, read_all_from_fd, pw_timing_setup, prepare_extension	Before compilation	Call only	Return or process exit	None beyond existing pointers/rights	No: exec replaces stack	P	`parse_args`; `map_region`; `read_all_from_fd`; `pw_timing_setup`; `prepare_extension`	Source only; no assertion cited	
n9b	Capture hash/view/ digest/sorting call frames	reporting	N9b	pw_capture_profile, pw_consumed_params_digest, pw_read_memory	Between compilation and apply	Call only	Return or process exit	Read own memory with existing task right	No: exec replaces stack	P	`pw_capture_profile`	Source only; no assertion cited	
n10	Attempt-local stack buffers, argv pointers, polls, wait status and metadata	mixed	N10	run_attempt and each attempt_*, capture_observed_path, drain_one, exec_resources_acquire; attempt_sysctl_read holds a sysctl value-read buffer of 4,096 bytes on the stack	After apply within an attempt	Call/attempt only; selected observations copied into S18; sysctl value bytes and post-call length discarded	Return or process exit	Use already inventoried target/ child handles; no separate authority	argv_local strings copied; stack not inherited	O + P (working/ observation storage)	`run_attempt`; `attempt_sysctl_read`	`check_capacity` — Independent call receipt requires the fixed supplied sysctl capacity; a shortened-capacity mutant is rejected.; `check_call_and_result` — Substituted sysctl success/failure returns preserve rc/ errno semantics despite changed out-parameter length.	`sysctl_read_bytes`
n11	Post-apply outside-attempt automatic diagnostics and seam timespec	reporting	N11	wait_for_proceed, proceed_refused, W post-phase seam	After apply outside an attempt	Call, or nonreturning refusal frame until exit	Return/exit; not dynamically allocated by PW	None; diagnostic/ control working storage	No child spawned on these paths	P	`wait_for_proceed`	Source only; no assertion cited	

Id	Node	Kind	Inventory ID	Acquired by	When	Held across	Release / teardown action	Authority	Reaches exec children	Role	Sources	Checks and assertion scope	Limits
n12	Exec child stdin /dev/null fd 0	reporting	N12	exec_resources_acquire registers addopen; native spawn executes it	After apply within an attempt	Child lifetime	Child closes it or exits	Read EOF without exposing policy pipe	Yes, explicit child fd 0	P	`exec_resources_acquire`	`exec_pipes_per_attempt` — Repeated successful execs and surrounding reads under a hard descriptor ceiling constrain descriptor accumulation across attempts; call placement remains source-derived.; `contaminated_worker` — Independent observer receives the canary descriptor/environment in the worker adapter and excludes them from exec children; checks child stdio and stdin EOF.	
n13	Libc diagnostic state reachable after apply	reporting	N13	fail_from_errno, exec_setup_failed, attempt_exec_spawn, attempt_mach_lookup, run_attempt: snprintf, strerror	After apply within an attempt	Library-dependent; ordinary errno table is static, fallback buffer retained	Library/exit; no PW-owned handle	Diagnostic formatting; hidden allocation/locale behavior bounded only by source studied	Library memory not carried across exec	P	`fail_from_errno`; `exec_setup_failed`; `attempt_exec_spawn`; `attempt_mach_lookup`; `run_attempt`	Source only; no assertion cited	
x1	Validator predictions	external	X1	planValidatorQueries	Query collection outside worker	Eligible phase and reply	Owning host/controller object lifetime	Live worker PID query through separate validator	No worker-to-child transfer	External evidence	`planValidatorQueries`	`comparison_matrix` — Independent reader checks query/attempt association and phase-order evidence on the submitted comparison specimens.	
x2	Host disposition and comparison	external	X2	comparisonEvidence	After collection/publication	Reply	Owning host/controller object lifetime	Join worker publications, query records and process observations	No worker-to-child transfer	External evidence	`comparisonEvidence`	`comparison_matrix` — Independent reader checks query/attempt association and phase-order evidence on the submitted comparison specimens.	
x3	Host path diagnostics	external	X3	enrichPathDiagnostics	After attempt collection	Reply	Owning host/controller object lifetime	Later path/metadata observations under host authority	No worker-to-child transfer	External evidence	`enrichPathDiagnostics`	`path_query_integrated` — Integrated path-query controls check later host path resolution and its separation from the worker observation.	
x5	Compiler receipt in dossier	external	X5	decodeProfileCapture	Capture before apply; host decode after publication	Reply and optional dossier	Owning host/controller object lifetime	Copy/hash admitted receipt from shared evidence	No worker-to-child transfer	External evidence	`decodeProfileCapture`	`max_targets_reply_survives` — The maximal-metadata workload requires a captured compiled-profile receipt with positive bytecode length in the returned reply.	

Worker capabilities edges

Id	From	To	Kind	Edge	Sources	Checks and assertion scope	Limits
W1	s1b	s2	prepares	Host maps evidence descriptor	`runCWorker`	`check` — Independent mapping challenges distinguish the true evidence object; worker reporting and final reap remain intact.	

Id	From	To	Kind	Edge	Sources	Checks and assertion scope	Limits
W2	s1a	s1b	prepares	Host holds backing-object descriptor through worker launch	`runCWorker`	`check` — Independent object, descriptor-number reuse and complete-mapping teardown controls; successful worker reporting and reap remain intact.	
W3	s1b	s13b	prepares	Spawn action duplicates evidence descriptor	`runCWorker`	`check` — Independent mapping challenges distinguish the true evidence object; worker reporting and final reap remain intact.	
W4	s13b	s18	prepares	Worker maps evidence descriptor	`runCWorker`; `map_region`	Source only; no assertion cited	
W5	s3b	s13c	prepares	Spawn action duplicates readiness endpoint	`runCWorker`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
W6	s4a	s13a	prepares	Spawn action duplicates policy endpoint	`runCWorker`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
W7	s4b	s23	prepares	Policy delivery fills static source buffer	`runCWorker`	Source only; no assertion cited	
W8	s6	s7	prepares	Worker launch file actions select protocol endpoints	`runCWorker`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
W9	n5	s7	prepares	Copied argv supplies worker launch arguments	`withCStringArrayCopy`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
W10	n6	s2	prepares	Host initializes request fields and nonce	`runCWorker`	Source only; no assertion cited	
W11	s23	s26a	prepares	Compiler reads source buffer	`main`	`bare_deny_default` — The direct C harness observes applied policy, terminal publication and a denied file-open attempt under bare deny-default; host Swift handling is outside this control.	
W12	s24	s25	prepares	Snapshot entries populate compiler parameters	`main`	`params_round_trip` — Named TARGET parameter reaches the compiled policy: the configured path attempt returns EPERM after successful apply; allocation/free details remain source-only.	
W13	s25	s26a	prepares	Compiler receives parameter object	`main`	`params_round_trip` — Named TARGET parameter reaches the compiled policy: the configured path attempt returns EPERM after successful apply; allocation/free details remain source-only.	
W14	s26a	s27	prepares	Capture reads returned compiled object	`main`; `pw_capture_profile`	`max_targets_reply_survives` — The maximal-metadata workload requires a captured compiled-profile receipt with positive bytecode length in the returned reply.	
W15	s35a	s36a	prepares	Spawn file actions select child descriptors	`exec_resources_acquire`	`exec_pipes_per_attempt` — Repeated successful execs and surrounding reads under a hard descriptor ceiling constrain descriptor accumulation across attempts; call placement remains source-derived.; `contaminated_worker` — Independent observer receives the canary descriptor/environment in the worker adapter and excludes them from exec children; checks child stdio and stdin EOF.	

Id	From	To	Kind	Edge	Sources	Checks and assertion scope	Limits
W16	s35b	s36b	prepares	Spawn attributes select child process group	`exec_resources_acquire`	`closed_streams_leader_exit` — Independent peer identity/group observations establish same-group survival after ordinary leader-plus-EOF completion, then fixture cleanup.	
W17	n2	s9	prepares	Issuer file actions select channel and stdio	`ExtensionSession`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
W18	n3	s9	prepares	Issuer attributes select descriptor isolation	`ExtensionSession`	Source only; no assertion cited	
W19	s8b	n1	supplies_authority	High duplicate of session descriptor	`ExtensionSession`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
W20	n1	s10c	supplies_authority	Issuer receives channel fd	`ExtensionSession`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
W21	s10c	s10b	supplies_authority	Issuer maps channel fd	`ExtensionSession`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
W22	s8a	s8b	supplies_authority	Session holds channel object descriptor	`ExtensionSession`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
W23	s8b	s8c	supplies_authority	Host maps authority-bearing channel	`ExtensionSession`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
W24	s8b	s5	supplies_authority	Host duplicates channel for worker delivery	`runCWorker`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
W25	s5	s13d	supplies_authority	Worker receives optional token descriptor	`runCWorker`	`check` — Setup controls compare requested dup2 destinations with child descriptor metadata, check argv counts and instrumented cleanup; shared-object identity is not established by metadata.	
W26	s13d	s21	supplies_authority	Worker maps token descriptor	`runCWorker`; `prepare_extension`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
W27	s11a	s11b	prepares	Loaded API symbols permit issuance	`main`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
W28	s11b	s12a	exercises	Native issuer returns bearer allocation	`main`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
W29	s12a	s12b	supplies_authority	Issuer copies accepted bearer bytes	`main`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	

Id	From	To	Kind	Edge	Sources	Checks and assertion scope	Limits
W30	s12b	s22	supplies_authority	Worker borrows same bytes through its mapping	`prepare_extension`; `attempt_extension`	Source only; no assertion cited	
W31	s22	s37	supplies_authority	Requested consume yields native extension handle	`prepare_extension`; `attempt_extension`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
W32	s26a	s28	exercises	Apply receives compiled object and imposes policy	`main`	`bare_deny_default` — The direct C harness observes applied policy, terminal publication and a denied file-open attempt under bare deny-default; host Swift handling is outside this control.	
W33	s33a	s33b	exercises	Lookup uses acquired bootstrap reference	`attempt_mach_lookup`	Source only; no assertion cited	
W34	s31a	s31b	exercises	Successful create-open can create absent file	`attempt_file_create`	`create_allow` — An initially absent target exists after the successful create attempt under allow-default; the worker also publishes its observed path.	
W35	s34b	s36a	exercises	Child receives output endpoints as stdout/stderr	`exec_resources_acquire`	`exec_pipes_per_attempt` — Repeated successful execs and surrounding reads under a hard descriptor ceiling constrain descriptor accumulation across attempts; call placement remains source-derived.; `contaminated_worker` — Independent observer receives the canary descriptor/ environment in the worker adapter and excludes them from exec children; checks child stdio and stdin EOF.	
W36	n12	s36a	exercises	Child receives explicit stdin	`exec_resources_acquire`	`exec_pipes_per_attempt` — Repeated successful execs and surrounding reads under a hard descriptor ceiling constrain descriptor accumulation across attempts; call placement remains source-derived.; `contaminated_worker` — Independent observer receives the canary descriptor/ environment in the worker adapter and excludes them from exec children; checks child stdio and stdin EOF.	
W37	s37	s32	exercises	Fresh read attempts access under current extension state	`attempt_extension`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
W38	s27	s18	publishes	Capture writes receipt into evidence mapping	`pw_capture_profile`	`max_targets_reply_survives` — The maximal-metadata workload requires a captured compiled-profile receipt with positive bytecode length in the returned reply.	
W39	s29	s18	publishes	Publishes open status and available observed path	`attempt_file_open_read`	`bare_deny_default` — The direct C harness observes applied policy, terminal publication and a denied file-open attempt under bare deny-default; host Swift handling is outside this control.	
W40	s30	s18	publishes	Publishes open status; write return discarded	`attempt_file_open_write`	`file_actions_have_exact_effects` — Independent file bytes establish the open-write/truncation effect; the worker does not publish the write return.	
W41	s31a	s18	publishes	Publishes file-create attempt status from open result	`attempt_file_create`	`create_existing_file_preserves_contents` — Pre-existing targets retain their bytes after create-open; this does not exercise absent-target creation.	
W42	s32	s18	publishes	Publishes native open/stat/read/close observations	`attempt_file_read_fresh`	`extensions` — Fresh-read and extension action evidence is compared with independent file bytes and native-call controls; native release is not a revocation assertion.	

Id	From	To	Kind	Edge	Sources	Checks and assertion scope	Limits
W43	s33b	s18	publishes	Publishes lookup status, not retained port authority	`attempt_mach_lookup`	`prediction_unavailable_attempt_observed` — The live local-name specimen retains native lookup attempt status; no right-count or deallocation assertion.	
W44	s36a	s18	publishes	Publishes spawn, wait and cleanup observations	`attempt_exec_spawn`	`exec_pipes_per_attempt` — Repeated successful execs and surrounding reads under a hard descriptor ceiling constrain descriptor accumulation across attempts; call placement remains source-derived.; `contaminated_worker` — Independent observer receives the canary descriptor/environment in the worker adapter and excludes them from exec children; checks child stdio and stdin EOF.	
W45	s37	s18	publishes	Publishes consume/release native result	`attempt_extension`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
W46	s10b	s8c	publishes	Issuer publishes through same channel object	`main`; `ExtensionSession`	`extension_boundaries` — Controlled issuer/worker native-call boundaries, delivery/refusal metadata and token confinement; private library allocation internals remain source-only.	
W47	s3a	s13c	observes	Host reads worker readiness byte	`runCWorker`	`runCWorkerLifecycleTests` — Controlled worker readiness, release/acknowledgement and exit/reap observations.	
W48	s2	s18	observes	Host observes publications in its retained evidence mapping, including after worker exit	`runCWorker`	`runCWorkerLifecycleTests` — Controlled worker publications, ownership, acknowledgement reconciliation and confirmed reap; includes acknowledgement recovered after abnormal exit.	
W49	s34a	s36a	observes	Worker drains child output	`attempt_exec_spawn`; `drain_one`	`exec_pipes_per_attempt` — Repeated successful execs and surrounding reads under a hard descriptor ceiling constrain descriptor accumulation across attempts; call placement remains source-derived.; `contaminated_worker` — Independent observer receives the canary descriptor/environment in the worker adapter and excludes them from exec children; checks child stdio and stdin EOF.	
W51	s2	x2	joins	Host uses admitted publications	`runCWorker`; `comparisonEvidence`; `observeAcknowledgement`	`runCWorkerLifecycleTests` — Controlled worker publications, ownership, acknowledgement reconciliation and confirmed reap; includes acknowledgement recovered after abnormal exit.	
W52	x1	x2	joins	Host joins prediction records to attempt evidence	`planValidatorQueries`	`comparison_matrix` — Independent reader checks query/attempt association and phase-order evidence on the submitted comparison specimens.	
W53	x3	x2	joins	Host attaches later path-resolution results	`enrichPathDiagnostics`	`path_query_integrated` — Integrated path-query controls check later host path resolution and its separation from the worker observation.	
W54	s27	x5	joins	Host decodes and copies receipt	`pw_capture_profile`; `decodeProfileCapture`	`max_targets_reply_survives` — The maximal-metadata workload requires a captured compiled-profile receipt with positive bytecode length in the returned reply.	

Id	From	To	Kind	Edge	Sources	Checks and assertion scope	Limits
W55	s32w	s18	publishes	Publishes separate write and metadata observations through completed slot	`attempt_file_write_fresh`	`fresh_write` — Production syscall substitutions assert open/stat/truncate/write/close order, native failures, short writes and combined-failure preservation; no native policy effect is inferred.; `extension_directory` — Owned existing files and initially absent children provide independent contents and entry observations through ordinary CLI phases, with authority controls and publication withholding.	
W56	s32c	s18	publishes	Publishes separate write and metadata observations through completed slot	`attempt_file_write_fresh`	`fresh_write` — Production syscall substitutions assert open/stat/truncate/write/close order, native failures, short writes and combined-failure preservation; no native policy effect is inferred.; `extension_directory` — Owned existing files and initially absent children provide independent contents and entry observations through ordinary CLI phases, with authority controls and publication withholding.	

Notes:

- `s5`: Setup controls observe endpoint-action failure refusing spawn with cleanup.
- `s6`: Worker setup controls observe failed file-action initialization/addition refusing spawn without use of an invalid handle.
- `s12a`: Private issuance allocation internals remain unobserved.
- `s14a`: L observed read/write null-device stdout and stderr.
- `s14b`: L observed no additional numeric descriptor; the contaminated-worker control separately supplies a positive inheritance case.
- `s14c`: L observed no original-number evidence alias in its endpoint arrangement; source permits other arrangements.
- `s15`: L produced identical plain and environment-displaying ps output; this is scoped to those two launches.
- `s16a`: Selected signing metadata and required-key checks do not measure effective worker authority.
- `s16g`: L observed cwd / in both launches.
- `s16h`: No live umask measurement is credited.
- `s16c`: No live resource-limit census is credited.
- `s16i`: No live inherited-signal census is credited.
- `s16j`: No live signal-mask measurement is credited.
- `s16e`: L port inspection was refused; the exact live right set remains unknown.
- `s16f`: L retained vmmap output; per-allocation origin and policy gating remain unknown.
- `s17`: L pre-ready launch retained readiness errno EPIPE and later clean completion.
- `s19`: Private loader/library internals remain unobserved.
- `s21`: Rejected-mapping unmap failure is recorded on the preparation path.
- `s25`: Private parameter-allocation internals remain unobserved.
- `s26a`: Private compiled-object lifetime inside apply remains unknown.
- `s33a`: No independent right-count control is cited for the retained bootstrap reference.
- `s33b`: No independent right-count control is cited for service-right release.
- `s35a`: The examined Apple source uses malloc/realloc; installed allocation behavior is unobserved.
- `s35b`: The examined Apple source uses malloc; installed allocation behavior is unobserved.
- `n2`: Failed initialization reaches fixture-contained invalid-handle use in setup controls.
- `n3`: Failed initialization reaches fixture-contained invalid-handle use in setup controls.
- `n4`: Issuer stdio file actions have source support only; extension boundary controls do not assert these endpoint identities.
- `n5`: Controlled strdup failure can dispatch empty or truncated argv.
- `n6`: Swift allocation internals and temporary-array lifetimes have source support only.
- `n9a`: Native library internals remain unobserved.
- `n9b`: qsort/CommonCrypto internal allocation behavior remains unobserved.
- `n10`: Compiler stack/page behavior remains unobserved; the sysctl checks cover supplied capacity and return handling.
- `n13`: Installed libc allocation and locale internals remain unobserved.
- `x1`: No transfer into worker; release ordinal is separate
- `x2`: Host-owned interpretation under evidence contract
- `x5`: Receipt and applied-policy evidence are distinct
- `W4`: Source identifies the descriptor-to-mapping operation. L does not independently identify the evidence mapping object.

- `W51`: A final acquire after confirmed reap can recover an unseen acknowledgement while ownership remained unbroken; successful completion is not required and broken ownership cannot be repaired.

Claims without a cited check: `s14c`, `s16a`, `s16h`, `s16c`, `s16i`, `s16j`, `s18`, `s33a`, `s33b`, `s35a`, `s35b`, `n4`, `n6`, `n8`, `n9a`, `n9b`, `n11`, `n13`, `W4`, `W7`, `W10`, `W18`, `W30`, `W33`.

Known gaps

- Starting authority and ambient context: the launch observation is a bounded census of inherited and runtime state.
- Acquisitions inside attempts: exec descendants can survive ordinary completion or worker death; escaped descendants are not contained.
- Supplied extension authority: private native state and allocation extend beyond PW-owned buffer and publication observations.